

OUTLINE

- Machine Learning & Data Engineer with 4 years of industry experience. Former Immunology Scientist with 10 years of research expertise. Passionate about leveraging technology to create meaningful impact. Driven by work that improves people's lives. Mom worked as a cleaning lady. Dad was a plumber. They taught me grit, optimism and work ethic.

EXPERIENCE

- **Senior Biotechnology Research Officer** Jul 2024 - Sept 2025
International Centre for Future Generations, Biotech / AI / Policy *Brussels, Belgium*
 - **Impact:** Drive strategic biotechnology policy through research analysis and AI implementation at leading think tank.
 - * Analyze cutting-edge biotech research to inform global policy decisions
 - * Forecast biotechnology and AI trends affecting biosecurity and public health
 - * Built and deployed AI pipeline for automated knowledge database management (Python, LLM APIs)
- **MLOps Engineer (Contractor)** Jan 2024 - Jun 2024
AXA Belgium / AI / AWS Cloud *Brussels, Belgium*
 - **Achievements:** Accelerated AI deployment and modernized infrastructure for Belgium's leading insurer.
 - * Developed AWS bucket management application using Boto3 and Python
 - * Modernized dependency management stack (uv, ruff, pre-commit hooks)
 - * Successfully deployed production LLM (Mistral) on AWS Lambda
 - * Migrated legacy deployment pipelines to AWS Python SDK
- **Data Scientist** Sep 2022 - Oct 2023
Sunrise, MedTech / Backend / MLOps *Namur, Belgium*
 - **Transformation:** Led complete technical transformation of ML-powered sleep apnea diagnostic platform.
 - * Refactored entire codebase with comprehensive testing (0% to 96% coverage)
 - * Built advanced sleep analysis algorithms (oxygen desaturation, hypoxic burden, AASM-compliant scoring)
 - * Scaled data processing using Google Cloud Functions and Cloud Run
 - * Implemented MLOps training platform on Vertex.ai with full tracking capabilities
 - * Created interactive Python dashboards for patient data visualization (Streamlit, Altair)
 - * Deployed modern ETL pipeline (Airbyte, Airflow, Apache Superset) for business insights
 - * Automated scientific report generation using literate programming (Papermill, Quarto)
 - * Established CI/CD deployment automation (Ansible, GitHub Actions)
- **Machine Learning & Data Engineer** Sep 2021 - August 2022
Biolizard, Biotech / Backend / Data / Cloud Engineering *Ghent, Belgium*
 - **Delivery:** Delivered enterprise-grade data solutions for life sciences digital transformation.
 - * Architected cloud-native data pipelines on AWS and GCP (Scala, Spark)
 - * Deployed MLOps platforms for genomics and single-cell analysis (Metaflow, MLFlow, Vertex.ai)
 - * Built end-to-end data analytics and visualization solutions
- **Data Engineer Trainee** Jan 2021 - Jul 2021
Biolizard, Biotech / Backend / Data / Cloud Engineering *Ghent, Belgium*
 - **Foundation:** Mastered enterprise data infrastructure and engineering best practices.
 - * Deployed production Hadoop clusters with full Kerberos security (Ansible automation)
 - * Implemented transactional data lake platforms (Delta Lake, Apache Hudi)
 - * Developed robust data pipelines on Databricks following software engineering standards

- **Academic Researcher** Jan 2011 - Jan 2020
Academia, Research / Immunology / Bioinformatics France, Germany and Portugal
 - **Excellence:** Led groundbreaking immunology research with computational approaches across top European institutions.
 - * Published 20+ peer-reviewed papers in high-impact journals
 - * Supervised 4 graduate students (3 Master's, 1 PhD)
 - * Secured multiple competitive international grants (Marie Curie, EMBO, FCT)
 - * Presented research at prestigious international conferences

EDUCATION

- **Applied Master of Data Engineering for Artificial Intelligence** Mar. 2020 – Jun. 2021
Data ScienceTech Institute Paris, France
- **PhD in Cellular and Molecular Aspects in Pathology** Aug. 2011 – Nov. 2014
Nantes University Nantes, France

SKILLS

- **Programming Languages::** Python, Scala (Spark), Ruby, Crystal
- **DevOps::** Ansible, GitHub Actions, Bitbucket Pipelines, Git, Docker, Testing (Unit/Integration), Agile/Scrum
- **Cloud::** Google Cloud (Vertex.ai, Cloud Run, Functions, Pub/Sub, DataProc), AWS (S3, Lambda, StepFunctions, SDK)
- **Backend::** Flask, FastAPI, Databases (PostgreSQL, MySQL, MongoDB, Redis)

HOBBIES

- **Sport:** Lifelong athlete - Karate (7 years), Rugby (12 years), Brazilian Jiu-Jitsu (since 2016). Believer in *mens sana in corpore sano*
- **Reading:** Science fiction enthusiast. Recent reads: “Story of Your Life” by Ted Chiang, “Les Furtifs” by Alain Damasio
- **Travel:** Experienced international professional with exposure to diverse cultures (Germany, Portugal, Belgium)

REFERENCES

- Cyril Le Mat - cyril.lemat@gmail.com, +33 6 86 22 05 43 (Sunrise)
- Andrea Del Cortona - andrea.delcortona@gmail.com, +32 470 25 55 67 (BioLizard)
- Antoine Magnan - a.magnan@hopital-foch.com, +33 6 34 53 21 20 (Academia)